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Analysis of the Diagenetic Changes in Complex Carbonates Using Rock Physics Algorithms

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Abstract

Carbonate reservoirs are prone to significant diagenetic changes. Diagenetic processes can interact with each other and either increase or reduce porosity and permeability. The shear modulus of the rock is not affected by the saturating fluid and hence provides an excellent tool for analyzing diagenetic changes such as dolomitization, cementation and karstification.

The proposed workflow combines inclusion theories with the multimineral analysis to assess mineralogical complexity. The accuracy of this analysis is crucial to characterize the effects of various diagenetic processes on elastic moduli of the effective media. The cementation of the grains is evaluated using contact theories based on the constant cement model. The inclusion theory allows to describe isolated vuggy pores and fractures as the spherical and elongated ellipsoid inclusions, respectively. The porosity of fractures is assessed using the total porosity versus shear modulus cross-plot and is identified as the points falling below certain threshold.

An analysis was carried out on the carbonate reservoir of complex mineralogy, which was affected by diagenetic changes. The workflow produced variable cementation curve over the reservoir interval. The analysis identifies intervals with the presence of dissolution pores and fractured intervals. In many cases, the vuggy pores are not well connected, and the presence of such pores leads to a significant permeability reduction. On the other hand, the presence of open or dissolution-enhanced fractures leads to permeability increase. The mineralogy should be taken into account in the diagenesis analysis because of the mineralogy-related variations in the matrix shear modulus. Thus, ignoring the mineralogy can lead to erroneous results.

The combination of the contact and inclusion models with the multimineral analysis allows to provide qualitative and quantitative estimation of various diagenetic processes, such as dolomitization, cementation and karstification. The combination of different models, based on the inclusions and contact theories, provides a novel approach to the analysis of the aforementioned processes.

Introduction

Carbonate rocks, depending on a depositional environment, can be deposited as big size bioclast, well sorted grains or mud deposits and are traditionally characterized by early diagenetic changes. These changes include early cementation, fracturing and leaching. Early cementation leads to the formation of stiff rocks soon after burial of deposited carbonate rocks due to a process of carbonate dissolution and its precipitation as cement material. At the same time, dissolution processes create dissolution or vuggy porosity. Stiffer, well cemented rocks are prone to fracturing. As a result, fractures are often observed in carbonate reservoirs. In many cases these fractures can be enhanced by dissolution.

Figure 1 shows thin sections of different types of the carbonate reservoirs. Figure 1a shows relatively well sorted grainstone, some cementation on the grain contacts can be observed. Figure 1b shows grainstone with the presence of bioclasts, some dissolution porosity (vugs) and minor fractures are observed within bioclasts. Figure 1c shows fractured carbonate reservoir, where some bioclasts are incorporated within mudstone. All three thin sections are coming from the same well within 50 meters long reservoir section. Thus, reservoir properties changes can be significant within relatively small section of the well. Such abrupt changes make permeability prediction extremely challenging in the carbonate reservoirs. As we can see in Figure 1, isolated dissolution vuggy porosity can be observed in the vicinity of the classical granular reservoir or fractured reservoir. As a matter of fact, all three porosity types, namely – intergranular, vuggy and fracture can be present within depth interval of one foot (vertical resolution for the density logging tool or two quants for the standard logging records). For the intergranular pores, Kozeny-Carman (Carman, 1937), Timur-Coates (Coates et al., 1999) or RQI-FZI (Amaefule et al., 1993) model works quite well for the permeability estimation. The presence of isolated vugs leads to lower permeabilities for the same porosity ranges, when comparing to the intervals with intergranular porosity. On the other hand, the presence of fractures leads to a significant increase in permeability even for the intervals with relatively low porosity. Chen et al. (2008) introduced modifications to the Timur-Coates equation to account for isolated vugs. Fracture permeability can be estimated as the function of fracture porosity and fracture aperture (Dicman et al., 2004). As it was discussed in Vorobiev et al. (2013), intergranular and vuggy matrix permeability can be reconciled using dominant permeability estimate.

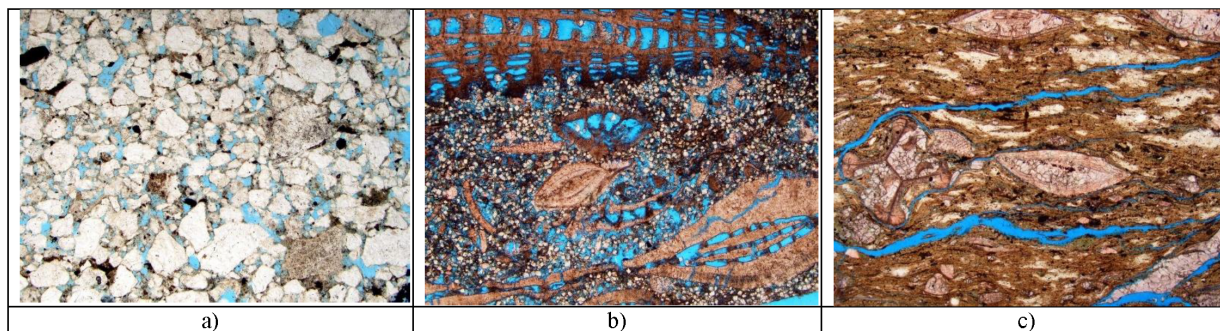


Figure 1—Thin section examples of different carbonate reservoirs: (a) grainstone; (b) grainstone with bioclasts and vuggy porosity; (c) mudstone with bioclasts in the presence of fractures.

As it can be seen, one of the major challenges is to estimate fracture, vuggy and intergranular porosity. It is especially difficult in cases with limited logging suite. The most common technique used for the quantitative evaluation of the vuggy and fractured porosity is based on the image logs binarization. Obviously, only visible fractures and vugs can be evaluated using image logs binarization.

In this paper we will review estimation of fractured, intergranular and vuggy porosity using Gamma Ray, Density, Neutron, Compressional and Shear Sonic logging suite. The technique utilizes rock physics modeling and was introduced earlier in Vorobiev et al. (2013) and Vorobiev et al. (2015). Important modifications were introduced to that technique. Earlier modification assumes that reservoir

with intergranular porosity follows friable sand model (Prasad and Dvorkin, 2001) and, hence, assumes no grains cementation. As thin section shows, some cementation is expected for the intergranular reservoir. It will effectively lead to the increase of both dry shear and dry bulk moduli of the frame and, as a result, to the artificially higher vuggy porosity and lower fracture porosity. In the updated version of the workflow, cemented sand model (Dvorkin et al., 1996) is used to define base intergranular trend in terms of the shear moduli.

When comparing image logs-based method with the shear sonic based one, obviously, image logs are the direct observations of either fractures or vugs, while sonic is measuring indirect response to the changes in pore geometry. On the other hand, only major vugs and fractures are visible on the image logs, while sonic is sensitive to both macro- and micro- fractures or vugs. The method has its limitations, which will be discussed in detail in the Conclusions section.

In the paper well F17-12 was used for the analysis. Well data is available in the Netherlands opensource database (<https://www.nlog.nl/datacenter/>).

Methodology

As we can see in the Figure 1a, intergranular carbonate reservoir is quite similar to the clastic one in texture with only two important exceptions. These are:

- Carbonate reservoirs quite often have complex matrix and consist of different minerals such as calcite, dolomite, quartz etc. Thus, proper modeling of the complex lithology requires pre-estimation of these mineralogical volumes and subsequent mixing of minerals in order to estimate matrix elastic moduli of the mix. Different bounds or, in some cases, inclusion models can be used for these minerals mixing procedure. The review of optimal procedures for the minerals' mixing is outside of the scope of this paper.
- As it was mentioned before, due to high solubility of calcite, cementation starts at an early stage. Thus, it is unlikely that friable sand model will be applicable for such types of reservoirs. Using cemented sands model is more viable.

As one can see in the Figure 1b, the majority of vugs are relatively large semicircular or rectangular isolated pores incorporated into solid media. In terms of elastic moduli such pores can be described as spherical inclusions into solid media. As we assume that these pores are disconnected from each other, Kuster-Toksoz spherical inclusions model (Kuster et al., 1974) fits well in this case.

Similarly, as one can see in the Figure 1c, fracture is a kind of elongated pore with significant extent in comparison to the width. Such pores can be described as well with the Kuster-Toksoz model using penny-crack inclusions. Penny-crack inclusions are a type of 2D elongated elliptical inclusions. The geometry of such inclusion is defined by the aspect ratio α of the axis of ellipse ($x=y$, $\alpha=z/x$).

Kuster-Toksoz model can be expressed as follows:

$$(K_{kt} - K_m) \frac{(K_m + \mu_m)}{(K_{kt} + \mu_m)} = \sum_{i=1}^n X_i (K_i - K_m) P_{mi} \quad (1)$$

$$(\mu_{kt} - \mu_m) \frac{(\mu_m + \varepsilon_m)}{(\mu_{kt} + \varepsilon_m)} = \sum_{i=1}^n X_i (\mu_i - \mu_m) Q_{mi} \quad (2)$$

Here K_{kt} and μ_{kt} are the bulk and shear moduli of the effective media with inclusions; K_m and μ_m are the bulk and shear moduli of the host media that, in our case, is carbonate. X_i is the volume occupied by inclusions and K_i and μ_i are the bulk and shear moduli of the inclusions, which are fluids in our case. Thus, in this case μ_i is equal to zero. Shear modulus ε_m is given by the equation below:

$$\varepsilon_m = \frac{\mu_m (9K_m + 8\mu_m)}{6(K_m + 2\mu_m)} \quad (3)$$

P_{mi} and Q_{mi} are the coefficients depending on the shape of inclusions. The formulas for different shape inclusions were derived by Berryman (Berryman, 2004). For the spherical inclusions coefficients can be written as:

$$P_{mi} = (K_m + \frac{4}{3}\mu_m) / (K_i + \frac{4}{3}\mu_m) \quad (4)$$

$$Q_{mi} = (\mu_m + \varepsilon_m) / (\mu_i + \varepsilon_m) \quad (5)$$

while for the "penny-crack" shape:

$$P_{mi} = (K_m + \frac{4}{3}\mu_i) / (K_i + \frac{4}{3}\mu_i + \pi\alpha\beta) \quad (6)$$

$$Q_{mi} = \frac{1}{5} \left(1 + 8\mu_m / (4\mu_i + \pi\alpha(\mu_m + 2\beta)) \right) + 2 \left(K_i + \frac{2}{3}(\mu_i + \mu_m) \right) / (K_i + \frac{4}{3}\mu_i + \pi\alpha\beta) \quad (7)$$

where:

$$\beta = \mu_m (3K_m + \mu_m) / (3K_m + 4\mu_m) \quad (8)$$

Base intergranular trend can be expressed using constant cement model:

$$K_{eff} = \left[\frac{\frac{PHIT}{PHIT_{cem}}}{K_{eff_cont} + \frac{4}{3}\mu_{eff_cont}} + \frac{1 - \frac{PHIT}{PHIT_{cem}}}{K_{min} + \frac{4}{3}\mu_{eff_cont}} \right]^{-1} - \frac{4}{3}\mu_{eff_cont} \quad (9)$$

$$\mu_{eff} = \left[\frac{\frac{PHIT}{PHIT_{cem}}}{\mu_{eff_cont} + \varepsilon_{eff_cont}} + \frac{1 - \frac{PHIT}{PHIT_{cem}}}{\mu_{min} + \varepsilon_{eff_cont}} \right]^{-1} - \varepsilon_{eff_cont} \quad (10)$$

where

$$\varepsilon_{eff_cont} = \frac{\mu_{eff_cont}}{6} \left(9K_{eff_cont} + 8\mu_{eff_cont} \right) / \left(K_{eff_cont} + 2\mu_{eff_cont} \right) \quad (11)$$

Cementation porosity ($PHIT_{cem}$) is related to the cement in pore space and critical cementation porosity as follows:

$$PHIT_{cem} = PHIT_{crit_cem} - cem \quad (12)$$

In general, constant cement model is interpolation between cementation porosity point ($PHIT_{cem}$) and zero porosity point. Low Hashin-Shtrikman model (Hashin and Shtrikman, 1963) is used to interpolate between these two end-points. Elastic moduli for zero-porosity point are equal to the mineral elastic moduli, or in the case of complex lithology to the effective moduli of the mix of the mineral moduli for the dry matrix. For the cementation porosity end-point elastic moduli are defined by contact cement model.

Figure 2 shows the principles of the proposed model. Constant sand model (grey solid line on the plot) defines base intergranular porosity line. Evaluation is done in terms of shear moduli as shear moduli are not affected by fluid effect, thus, if evaluation is done in terms of shear moduli only, the result will be affected by rocks texture only. When working just in elastic moduli terms, the algorithm can define either vuggy-intergranular domain or fracture-intergranular domain. Points falling above base line belong to the vuggy secondary porosity domain, while points falling below that line correspond to fractured secondary porosity domain. The solid green line on the plot shows elastic moduli behavior for the case of all pores being vugs. The line defined by Kuster-Toksoz model does the same for the spherical inclusions. The red solid line represents pure fracture porosity behavior and is defined by Kuster-Toksoz model for the penny-

crack inclusions. In the case when both intergranular and vuggy porosity exists (points above base line), the algorithm gradually increases vuggy porosity from zero to the value, which allows to match measured elastic moduli. Porosity in this case can be expressed as follows:

$$PHIT = PHI_{int} + PHI_{vug} \quad (13)$$

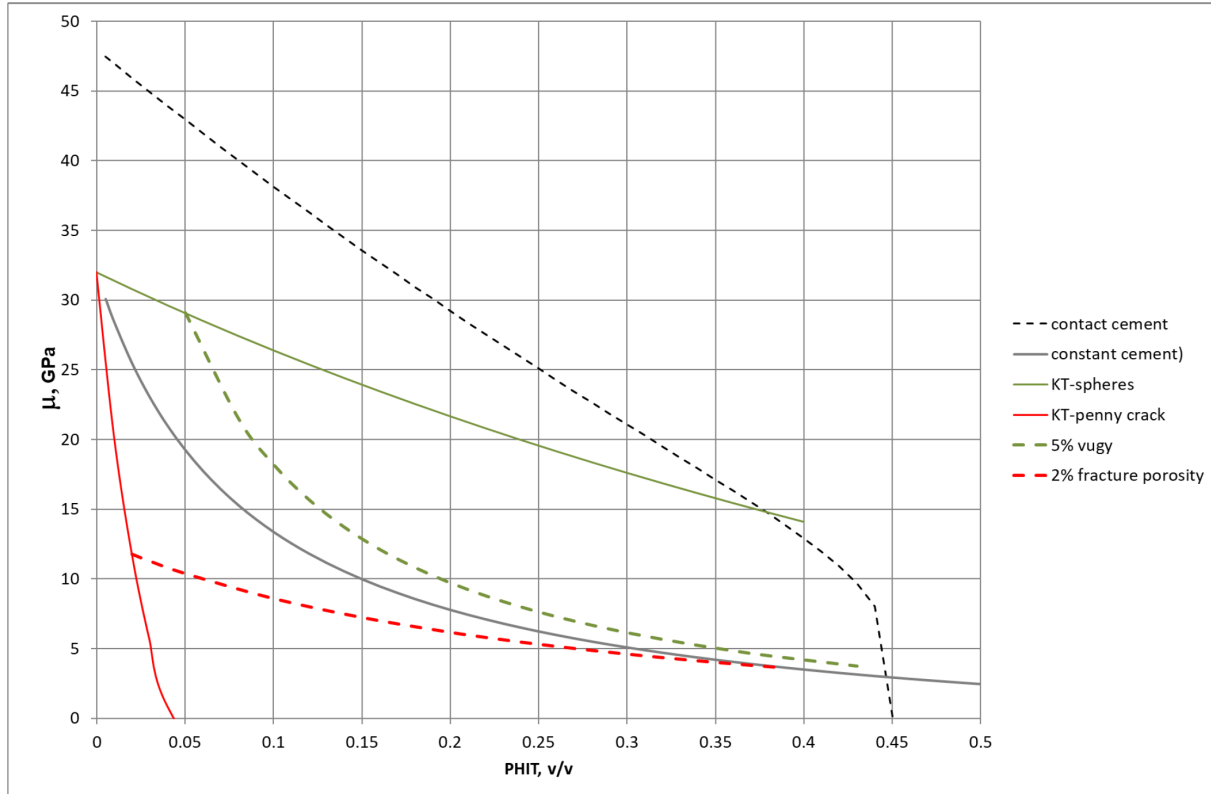


Figure 2—Model in terms of shear modulus.

In this case Kuster-Toksoz effective elastic moduli calculated for the selected PHI_{vug} value are used as the matrix moduli for the constant cement line, porosity in the constant cement line is reduced to intergranular porosity value (PHI_{int}). The green dashed line (Figure 2) shows the model behavior in the case of 5% vuggy porosity.

Similarly, in the case of the fracture domain, porosity can be expressed as follows:

$$PHIT = PHI_{int} + PHI_{frac} \quad (14)$$

The algorithm gradually increases fracture porosity PHI_{frac} and decreases intergranular porosity PHI_{int} . Similar to the vuggy branch, pseudo matrix elastic moduli are used as the input to the constant cement model. These moduli are calculated using Kuster-Toksoz model with penny-crack inclusions, the volume of the inclusions is defined by the fracture porosity PHI_{frac} . The 2% fractures porosity line is shown in Figure 2 as dashed red line.

Model implementation requires to identify average cementation for the reservoir. As it can be seen from Figure 2, increase in base cementation will shift the grey line upward. In this case for the same data point in total porosity ($PHIT$) and shear moduli coordinates, if the point falls above the constant cement line, selection of higher cement in pore space will effectively reduce vuggy porosity. Similarly, for the fracture domain, the selection of the higher cement in pore space for the base-line will lead to the higher fracture porosity (PHI_{frac}) for the same shear moduli-total porosity point.

Results

The well F17-12 was drilled through the chalk reservoir. The logging suite includes Gamma Ray, Resistivity, Neutron, Density, Compressional and Shear sonic, NMR and Wireline Formation Tester. Core samples, which are available in the database, indicate grain density variations 2.68-2.75 g/cc due to the presence of different minerals in addition to calcite (potentially chert and some other). Porosity and lithology were evaluated using Gamma Ray, Neutron and Density logs as the input. Evaluation results are displayed in Figure 3. Variable density (Neutron-Density cross plot) has been used to account for the grain density variability observed in core samples. Additionally, NMR total porosity was used for the model calibration. Lithology volumes, namely sand and dolomite in this case represent rather volumes with grain density lighter or heavier than calcite.

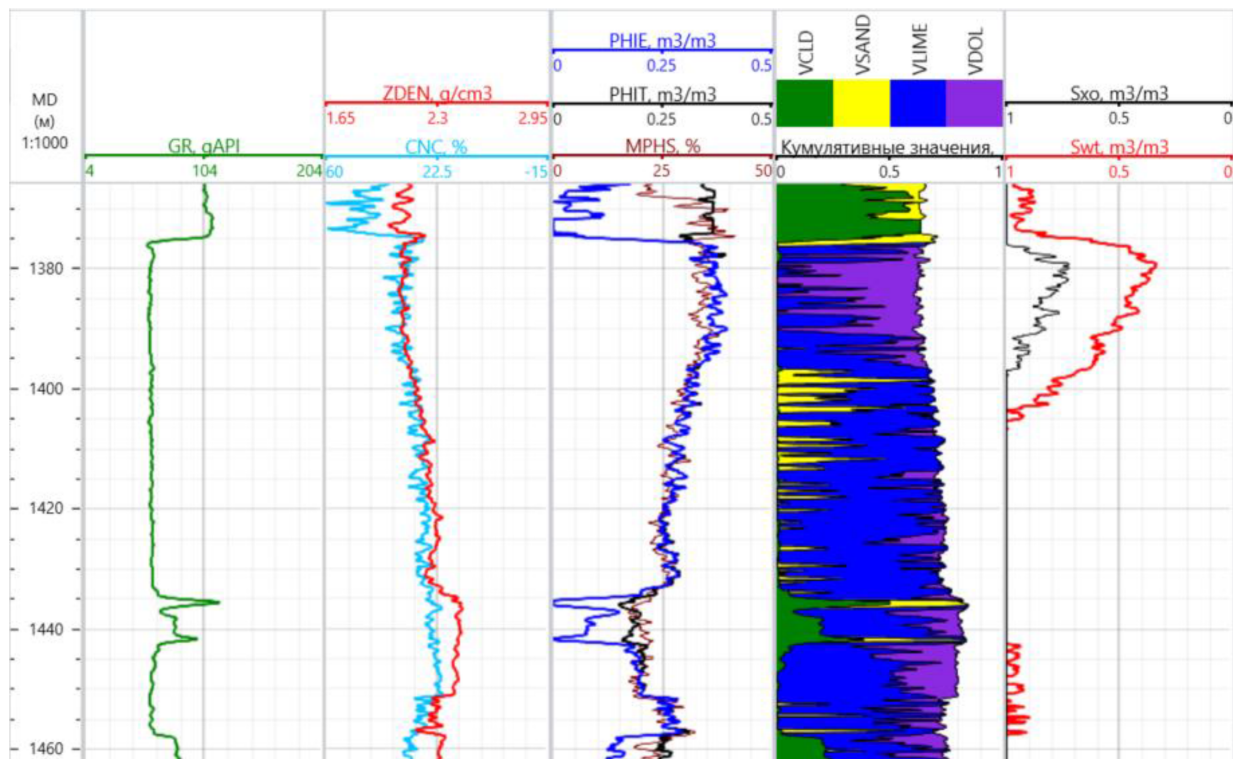


Figure 3—Well F17-12 lithology and porosity petrophysical evaluation results.

Figure 4 shows the plot of shear moduli as the function of total porosity. Data points are filtered by clay volume ($V_{clay} < 0.01$), so that lower elastic moduli of clay will not affect the plot. In general, the plot displays three major groups of points:

- The first one with porosities in range of 0.15-0.2 v/v and shear moduli in range of 6-8.5 GPa
- The second with porosity in range 0.23-0.37 v/v and shear moduli 4.4-7.2 GPa
- The third with porosity in range 0.25-0.4 v/v and shear moduli 5.2-8.1 GPa

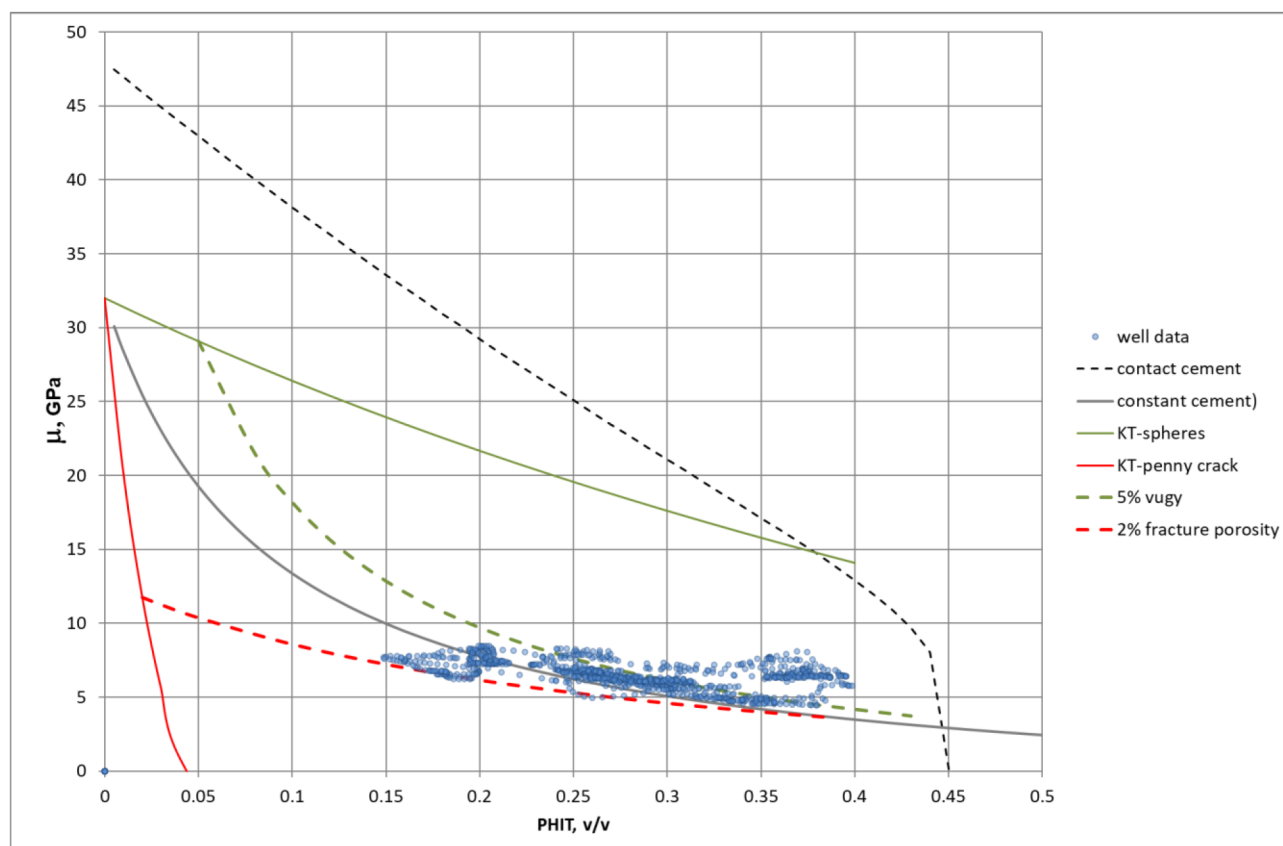


Figure 4—Shear moduli as the function of total porosity plot for the well F17-12 (chalk interval) with the model lines.

The last two trends are seen mostly as two parallel groups of data points, with quite similar porosity range. However, the third group has higher shear moduli. The first group of the data points concentrates in a lower porosity range.

If the constant cement line (grey solid line on the plot) is fitted into the bottom of the second group, the third group of points will fall below this base line. As mentioned above, such a behavior indicates the presence of fractures. In this case, the third group of points, characterized by higher shear moduli, must have higher percentage of vugs. Of course, these three trends can be explained alternatively as the group of rocks with the variable cementation. This scenario has been checked, by back-calculation of the cement in pore space parameter. This parameter is displayed on the last track in Figure 5. Figure 6 shows the same data points interpreted in terms of intergranular, vuggy and fractured porosity, where vuggy porosity is colored in orange in Track 4 and intergranular porosity is colored in cyan-blue in the same track. Track 5 displays the fractured porosity. As can be seen, the majority of the vuggy porosity falls into the same interval which can be interpreted as having higher cementation (the third group of points) and is located at the top of the reservoir (1377-1393 m MD). The first group of points, which can be interpreted as the presence of fractures, corresponds, as expected, to lower cementation derived from back-calculation of the cement volume in place and observed in the lower section of the reservoir (1435-1456 m MD). From the general geological perspective, the upper parts of the carbonate reservoirs are more prone to the karst and vugs' development. On the other hand, higher stress concentration at the base of reservoir, especially considering salt tectonic, makes base reservoir layers more prone to fracturing. Unfortunately, image logs are not available in this well to validate the evaluation. However, Wireline Formation Tester was run to estimate pore pressure and, as a result, WFT mobility data is available.

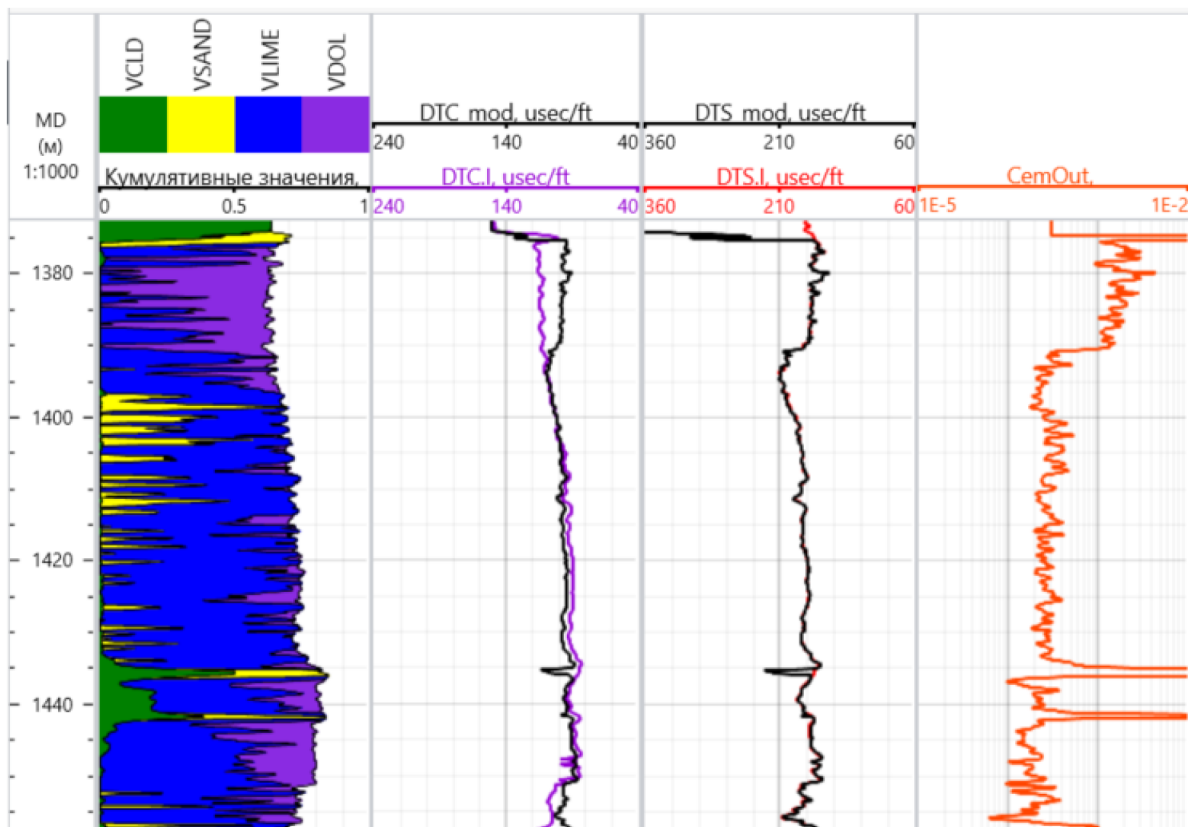


Figure 5—Well F17-12 run of the constant cement model with variable cementation estimation (only intergranular porosity assumed).

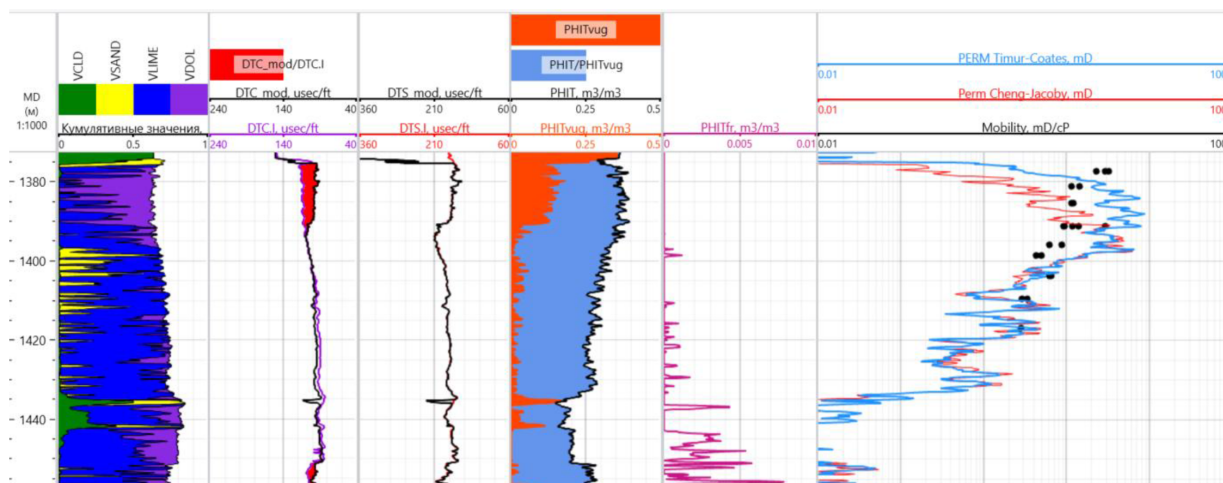


Figure 6—Well F17-12 evaluation of intergranular, fractured and vuggy porosity and consecutive evaluation of Chen-Jacoby permeability.

As mentioned previously, NMR run is available as well. NMR permeability was estimated using Timur-Coates equation and is displayed in the last track of Figure-6 (blue curve). As can be seen, with the exception of the first three WFT points taken at the top of the reservoir, the rest of the mobility points are lower than the Timur-Coates estimated permeability. In Chen et al. (2008), a modification of the Timur-Coates equation (Chen-Jacoby permeability equation) has been proposed:

$$k, mD = \left(\frac{\phi}{c} \right)^m \left(\frac{p \cdot BVM}{BVI + (1-p) \cdot BVM} \right)^n \quad (15)$$

The equation takes into account disconnected vugs. The term p represents disconnected porosity. The permeability evaluated by using Chen-Jacobi equation is displayed in Figure 6 in the Trak 6 (red curve). Figure 7 shows a comparison of the estimated permeability and WFT mobility. The permeability estimated by using Timur-Coates equation is in general higher than WFT mobility (Figure-7a) and the majority of the points falls below the one to one line. While for the Chen-Jacobi equation estimated permeability (Figure-7b) the points fall both above and below the one to one line.

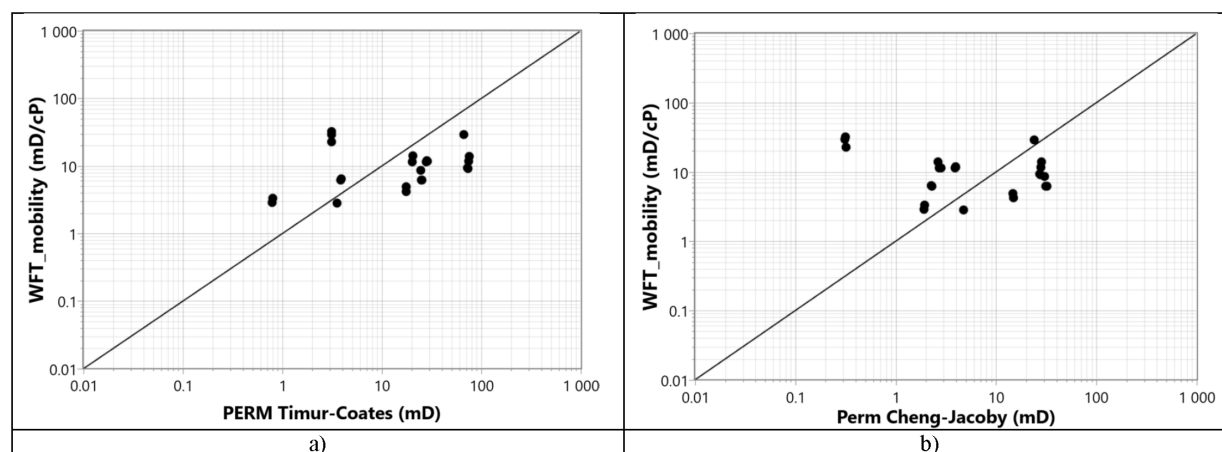


Figure 7—Comparison of the WFT mobility and permeability: (a) from Timur-Coates equation ignoring vuggy porosity; (b) from Cheng-Jacobi equation, considering vuggy porosity.

In some cases, it is possible to evaluate the saturation using compressional sonic. As it was mentioned earlier, the minimization is done in terms of shear moduli. The result is that the reconstructed shear sonic provides a close match for the measured one. However, one can observe a separation between the reconstructed and measured compressional slowness (Figure 6). The interval, where the measured compressional slowness is lower than the reconstructed one (1377-1393 m MD) contains hydrocarbons. The interval 1410-1423 m MD measured compressional slowness is higher than the reconstructed one, which indicates heavier mud filtrate invasion.

Conclusions

The proposed model allows to evaluate fracture and vuggy porosity in the case when limited logging suite is available. The minimum required logging suite for the evaluation is limited to Gamma Ray, Density, Neutron, Compressional and Shear slowness measurements and is available in majority of wells. The method does not allow to evaluate all three porosity components, namely intergranular, vuggy and fractured. For a single depth point the algorithm allows to evaluate either intergranular-fractured or intergranular-vuggy porosity. Variable cementation for the intergranular part of reservoir can mask the vuggy porosity effect, as both increase in cementation and increase in vuggy porosity leads to similar increase in shear moduli. As it was shown, the selection of the average cement in pore space for the interval can significantly affect the outcome. Fracture porosity evaluation also depends on the selection of the average aspect ratio for the fractures. The parameter can be calibrated with drill stem test (DST), especially if production logging tools (PLT) logs are available and it is possible to estimate each interval contribution to the flow.

Evaluation of the vuggy, fractured and intergranular porosity allows more accurate assessment of the permeability. One should remember that some vugs may not be disconnected, thus, in case if one treats all pores as disconnected, it may lead to lower permeability estimate.

As it was shown, it is possible to evaluate saturation by comparing the reconstructed and measured compressional slowness.

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